THE INTEGRAL OF A POSITIVE WEIGHTING IS LOWER SEMI-CONTINUOUS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 22702-22728, immediately following the statement at 22694-22700.

Editorial note. The proof cites the etale splitting lemma and uses the defining local constancy of a weighting; both are navigation pointers into the same chapter and are not re-counted.

Editorial note. The displayed chain of equalities for the integral over U is retained verbatim, including the textstyle integrals.

Allowed standard prerequisites (editorial).

- Etale splitting off of the quasi-finite part. chapter lemma lemma-etale-splits-off-quasi-finite-part-technical-variant, cited at proof lines 22704-22706. Gives the etale neighbourhood and the decomposition of U x_Y X into W and the V_{i,j} used to write the integral as a finite sum of locally constant pieces.
- Criterion for a weighting after etale base change. chapter lemma lemma-weighting-check-after-etale-base-change, cited via the definition of the integral. Explains why each integral over V_{i,j} of the restricted weighting is locally constant, hence constant after shrinking U.
- Base change compatibility of the integral. used at proof lines 22725-22726. Transfers the lower bound obtained over the etale neighbourhood back to Y.

Source reference. Original lemma 02LO: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12136–12136. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F39: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22111–22111. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a separated, locally quasi-finite morphism with finite fibres. Let $w : X \to \mathbf{Z}_{>0}$ be a positive weighting of f. Then $\int_f w$ is lower semi-continuous.*

Proof. Let $y \in Y$. Let $x_1, \ldots, x_r \in X$ be the points lying over y. Apply Lemma 02LO to get an étale neighbourhood $(U, u) \to (Y, y)$ and a decomposition

$$U \times_Y X = W \amalg \coprod_{i=1,\ldots,n} \coprod_{j=1,\ldots,m_i} V_{i,j}$$

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

as in locus citatus. Observe that $(\int_f w)(y) = \sum w(v_{i,j})$ where $w(v_{i,j}) = w(x_i)$. Since $\int_{V_{i,j} \to U} w|_{V_{i,j}}$ is locally constant by definition, we may after shrinking U assume these functions are constant with value $w(v_{i,j})$. We conclude that

$$\int_{U \times_Y X \to U} w|_{U \times_Y X} = \int_{W \to U} w|_W + \sum \int_{V_{i,j} \to U} w|_{V_{i,j}} = \int_{W \to U} w|_W + (\int_f w)(y)$$

This is $\geq (\int_f w)(y)$ and we conclude because $U \to Y$ is open and formation of the integral commutes with base change (Lemma 0F39). $\square$